

# Mayank Saxena

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## SUMMARY

Software engineer with 6+ years building and scaling security-critical distributed identity systems at Amazon (CloudAuth), serving 35,000+ customers. MS in Computer Science (Machine Learning track) from Columbia, with published research (AAAI 2018, ACM FAT\* 2018) and an IBM Watson patent in privacy-preserving ML. Now applying that systems expertise to research and engineering challenges in agentic AI, with a focus on security, identity, and authorization.

## EXPERIENCE

**Amazon** — New York, NY

Feb 2020 – Present

*Software Development Engineer I → Software Development Engineer II*

- **Core Contributor to CloudAuth:** Among the first 10 engineers on CloudAuth; built and evolved core authentication and authorization infrastructure as it scaled into Amazon's primary auth platform, serving 35,000+ customers and 100+ internal systems and supporting a 25x increase in request throughput over 5 years.
- **OAuth & Identity Protocols:** Implemented multiple OAuth 2.0 IETF RFCs and built transaction-token infrastructure for propagating workload identity across service call chains.
- **Security for Agentic AI:** Designed and built a permission-scoping framework extending that workload-identity and transaction-token infrastructure to agentic AI, enabling secure agent-to-agent and agent-to-service authorization for production AWS AgentCore workloads.
- **Global-Scale Privacy & Compliance:** Led cross-team integration of regional privacy requirements (GDPR, Quebec Law) into auth tokens, achieving global compliance with zero customer-facing disruptions.
- **Reliability & Mentorship:** Drove availability initiatives to exceed 99.99% SLAs, cut incident resolution time by 80%, and mentored new engineers as the team scaled 10x.

**IBM Thomas J. Watson Research Center** — New York, NY

Jun 2019 – Aug 2019

*Machine Learning Research Intern*

- **Encrypted Machine Learning:** Built scalable unsupervised learning algorithms for homomorphically encrypted data, enabling privacy-preserving analytics; filed a patent for a data-packing technique that improved clustering efficiency by 40%.

## PUBLICATIONS & PATENTS

- "Relating Children's Automatically Detected Facial Expressions to Their Behavior in RoboTutor," *Saxena, Pillai, Mostow* — **AAAI 2018**
- "Analyze, Detect and Remove Gender Stereotyping from Bollywood Movies," *Madaan, Mehta, Agrawaal, Malhotra, Aggarwal, Gupta, Saxena* — **ACM FAT\* 2018**
- **Patent filed (IBM, 2019):** data-packing technique for efficient clustering on homomorphically encrypted data
- *Plus 2 co-authored arXiv preprints (2017)*

## EDUCATION

**Columbia University** — New York, NY

Aug 2018 – Dec 2019

Master of Science, Computer Science (Machine Learning track) — GPA: 3.89/4.0

- Coursework: Machine Learning, Deep Learning, Computer Vision, NLP | TA: Visual Databases, Intro to Databases

**Delhi Technological University (Formerly DCE)** — New Delhi, India

May 2018

Bachelor of Technology, Mathematics and Computer Science — GPA: 3.66/4.0

## TECHNICAL SKILLS

- **ML/AI:** LLMs, Generative AI, Transformer Architectures, Reinforcement Learning, RLHF, NLP, Computer Vision
- **Languages & Frameworks:** Python, Java, C++, PyTorch, TensorFlow
- **Identity & Infrastructure:** OAuth 2.0, IETF RFCs, Transaction Tokens, Distributed Systems, AWS/AgentCore, Kubernetes

## SELECTED ML RESEARCH / PROJECTS

- **Spoiler-Sensitive Video Summarization:** RL-based approach to a novel spoiler-sensitive summarization objective.
- **Visual Question Answering System:** Alternating co-attention network for VQA, trained on multiple image-question datasets.